WEEK 2 REPORT - WEEKLY WORK

DG Interns Hub - SOC Analyst Internship

| Programme | DG Interns Hub - SOC Analyst Internship |

| Week | Week 2 - Networking, Log Types, Wireshark and MITRE ATT&CK |

| Submission deadline | 14 October 2026, 11:59 PM |

| Deliverables | 10 notes, 6 lab writeups, 6 hands-on investigations, this report, a presentation and a README |

Lab host	Windows host `EDITH` (192.168.0.205) and an Ubuntu VM

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1. Executive summary

Week 2 moved from the concepts of Week 1 to the evidence a SOC analyst actually works with: network traffic, log files and the MITRE ATT&CK framework.

The week produced three finished bodies of work:

1. Ten theory notes - one document per question, covering ATT&CK itself, tactics versus techniques, the three ATT&CK matrices, mapping an alert to a technique, ATT&CK versus the Cyber Kill Chain, protocol ports, DNS resolution, the DHCP DORA exchange, log formats and traffic direction.
2. Six TryHackMe room writeups - Network Traffic Basics, Intro to Logs, Intro to Log Analysis, Snort, DNS in Detail and HTTP in Detail, each documenting every task, the reasoning and the answer.
3. Six hands-on investigations - a packet-capture analysis, self-generated traffic, Linux log reading, a real Apache access-log analysis, an ATT&CK mapping exercise and a Snort IDS build.

All six hands-on tasks were completed against real artefacts rather than descriptions. The packet capture supplied with the brief was dissected packet-by-packet; a real 10,000-line public Apache `access.log` was analysed to answer the "top 3 IP addresses" question with exact counts; Snort 2.9.20 was installed and run to produce genuine alerts; and a real Ubuntu 22.04.5 LTS machine was booted so that the `auth.log` / `syslog` exercise could be performed on a real Linux system instead of described.

Headline finding of the week: the supplied capture contains no DNS and no HTTP-over-TCP traffic at all - it is an internal Windows SMB2 / MS RPC session plus an OpenVPN tunnel and SSDP discovery. That conclusion is proven with display filters and packet counts in `Findings Folder/1 Wireshark_Traffic_Capture_Analysis.pdf`, and it is a better lesson than a manufactured result: a capture only shows what was on the wire during its own window.

Second finding, from Wireshark's protocol hierarchy: the RPC traffic is not generic RPC. It carries Netlogon (126 frames) and DRSUAPI (52 frames) - the Active Directory replication interface behind DCSync. That is worth recognising because the same frames from a host with no business replicating the directory is a credential-theft incident.

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2. Objectives set by the brief, and how each was met

Objective	Status	Where
Answer all ten theory questions, one file each	Complete	`Notes/` - 10 documents

Objective	Status	Where
Complete the six free TryHackMe rooms and write up each	Complete	`Writeups/` - 6 documents
Capture and analyse the supplied traffic file	Complete	`Findings Folder/1 ...` + real Wireshark GUI screenshots and data images
Generate own traffic (`nslookup`, ping, download) and filter it	Complete	`Findings Folder/2 ...` + real command output and real Wireshark filter screenshots
Read `/var/log/auth.log` and `/var/log/syslog` for a sudo entry	Complete on a real Ubuntu 22.04.5 VM - 17 sudo lines, 5 `COMMAND=` entries	`Findings Folder/3 ...` + 6 real images
Analyse an Apache `access.log` and count the top 3 IPs	Complete with real data	`Findings Folder/4 ...` + 4 data images
Map three attacker behaviours to ATT&CK	Complete	`Findings Folder/5 ...` + 2 mapping images
Install Snort, test it, write a local rule and alert	Complete with real alerts - Snort 2.9.20 validated and run against 5 captures, 9 local rules	`Findings Folder/6 ...` + 7 real images
LinkedIn posts (5 points on ATT&CK, and a weekly recap)	Drafted	`LinkedIn Posts.md`
Weekly PDF report, presentation, README	Complete	This report, `Week 2 Presentation - Weekly Work.pptx`, `README.md`

3. Theory notes (`Notes/`)

Each note answers exactly one question from the brief, using bullet points and tables.

#	Document	Covers
01	What is MITRE ATT&CK?	Definition, origin, why it matters to a SOC, its building blocks (tactic, technique, sub-technique, procedure, group, software, mitigation, data source)
02	Tactics vs Techniques	Why = tactic, how = technique; a side-by-side comparison; 13 Enterprise tactics with an example technique each
03	ATT&CK Matrices	Enterprise, Mobile and ICS matrices, what each is for, and a comparison table of the three
04	Mapping Alerts to ATT&CK	The 8-step mapping method and a full worked scenario: Word spawning encoded PowerShell, mapped to T1059.001 with supporting techniques
05	ATT&CK vs Cyber Kill Chain	The seven kill-chain phases, a full comparison table, where the models overlap, and when to use each
06	Protocols and Default Ports	HTTP 80, HTTPS 443, DNS 53, DHCP 67/68, SSH 22, FTP 20/21, SMTP 25 (plus 587/465), with a Protocol / Port / Purpose table and encrypted-versus-plaintext analysis
07	How DNS Resolution Works	Every step from browser cache to authoritative nameserver, with the referral chain, record types and the SOC relevance
08	DHCP DORA	Discover, Offer, Request, Acknowledge explained step by step, with the message table, lease lifecycle and the rogue-DHCP threat

#	Document	Covers
09	Comparing Log Formats	Windows Event Log, syslog, JSON and CSV compared on structure, use case and example - plus the same event written in all four formats
10	Inbound vs Outbound Traffic	Definitions, a triage decision table, and how the distinction drives investigation and response

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4. Lab writeups (`Writeups/`)

Each writeup states the room and path, the objective, the concepts, and then every task with the question, the reasoning and the answer.

Room	Path	Tasks	Notable coverage
Network Traffic Basics	Cyber Security 101 / SOC L1	6	Why traffic analysis matters, fragmentation as IDS evasion, TCP sequence numbers and session hijacking, DNS tunnelling, and two flags recovered from HTTP and DNS traffic
Intro to Logs	SOC L1 - Log Analysis	7	Anatomy of a log entry, log types and formats, `rsyslog` collection, a reverse shell hidden in a cron job, `logrotate` retention, normalisation and enrichment
Intro to Log Analysis	SOC L1/L2 - Log Analysis	9	Super timelines, threat-intel indicators, path traversal in a URL, manual vs automated analysis, and a full command-line, regex and CyberChef investigation
Snort	SOC L1 - Snort module	11	IDS vs IPS families, all four Snort operation modes, PCAP investigation, and writing four local rules from scratch
DNS in Detail	Pre Security / Cyber Security 101	5	Domain hierarchy and its limits, all five record types, the full resolution path, and a practical record-hunting task
HTTP in Detail	Pre Security / Cyber Security 101	7	URLs, methods, status codes, headers, cookies, and five hands-on requests against a mock web server

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5. Hands-on investigations (`Findings Folder/`)

5.1 The supplied packet capture

The capture (`wireshark_capture.bin`, 204,849 bytes, pcap 2.4, Ethernet) was dissected and every figure in the findings document was measured from it.

Property	Value
Packets	1,171 (SHA-256 `88bb9d62...ff5b`)
Window	2020-09-20 21:21:36.455 UTC to 21:22:13.336 UTC (36.88 s)
Protocols	TCP 1,144 / UDP 25 / ARP 2
Source and destination IPs	192.168.100.128, 192.168.100.6, 192.168.100.1, 54.193.240.194, 239.255.255.250
Dominant service	SMB2 (113 frames) and MS RPC (423 DCE/RPC frames) between two internal hosts
AD-related RPC	Netlogon 126 frames, DRSUAPI 52 frames, endpoint mapper 64 frames
HTTP requests	None over TCP; 10 SSDP `M-SEARCH * HTTP/1.1` requests inside UDP 1900

Property	Value
DNS queries	None - `dns`, `udp.port == 53` and `tcp.port == 53` all return zero packets
Verdict	Normal internal file/RPC administration, an OpenVPN tunnel and UPnP discovery - no attack evidence

Wireshark Protocol Hierarchy - the supplied capture

REAL OUTPUT

Statistics > Protocol Hierarchy, produced by tshark 4.6.9 (real output)

Protocol layer	Frames	Bytes
frame	1171	186,089
eth	1171	186,089
ip	1169	185,987
udp	25	4,308
openvpn	15	2,370
ssdp	10	1,938
tcp	1144	181,679
dcerpc	381	114,208
epm	64	13,312
rpc_netlogon	126	16,346
drsuapi	52	25,236
nbss	115	27,833
smb	2	254
smb2	113	27,579
arp	2	102

This is what the capture actually contains - SMB2, DCERPC (Netlogon/DRSUAPI), OpenVPN and SSDP

Task 1 - Top IP Conversations

src -> dst pairs with packet and byte counts (Wireshark: Statistics > Conversations)

Source	Destination	Packets	Bytes	What it is
192.168.100.128	192.168.100.6	698	76497	SMB / RPC + dynamic-RPC traffic
192.168.100.6	192.168.100.128	446	105182	SMB / RPC replies (largest downloader)
192.168.100.1	239.255.255.250	10	1938	SSDP M-SEARCH discovery (UDP 1900)
54.193.240.194	192.168.100.128	8	1264	OpenVPN server -> client (UDP 1194)
192.168.100.128	54.193.240.194	7	1106	OpenVPN client -> server (UDP 1194)

Derived from the real capture - no packet content was invented

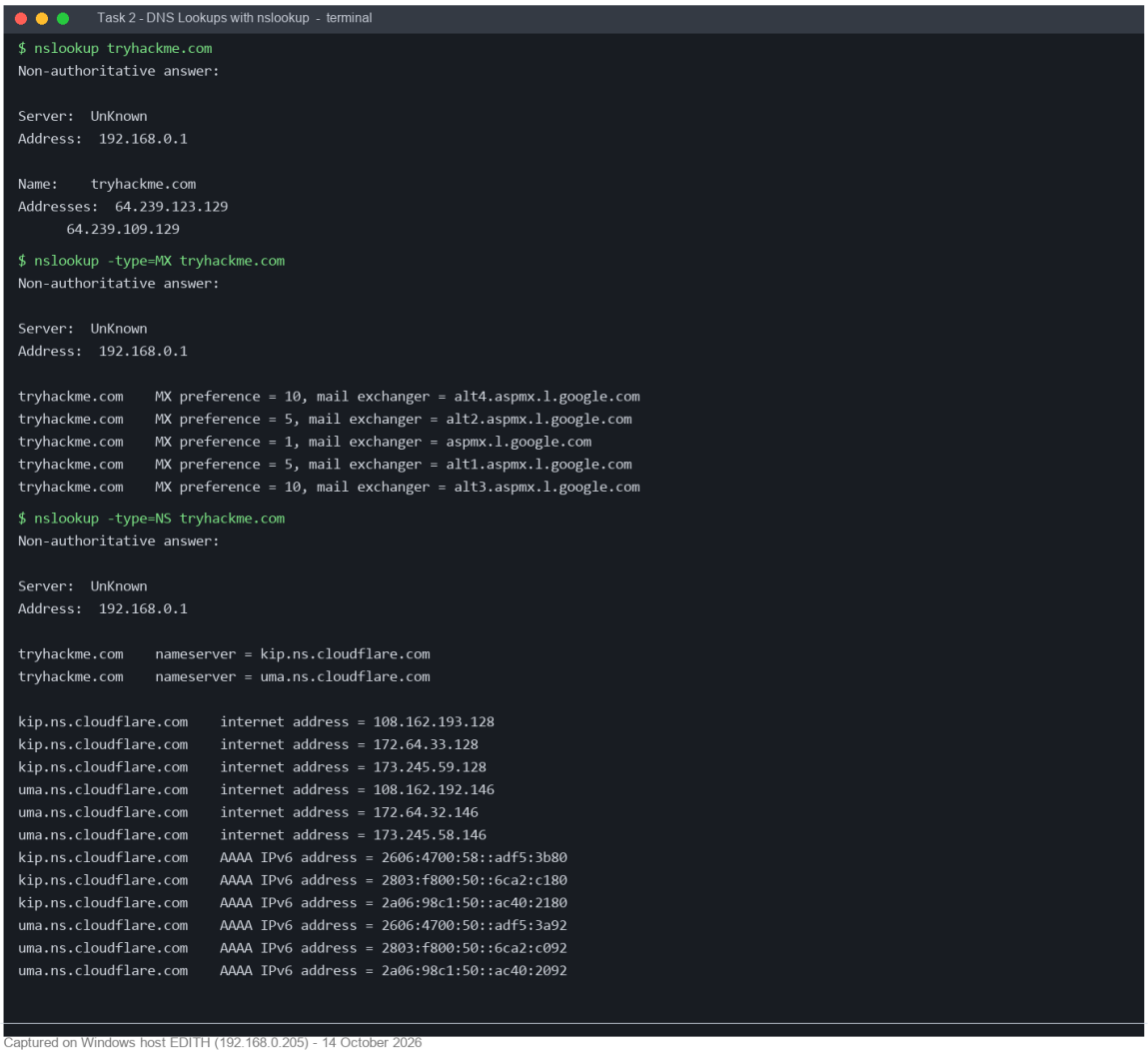
5.2 Self-generated traffic

Real DNS, ICMP, HTTP and DHCP evidence was captured on the host: `A`, `MX`, `TXT` and `NS` records for `tryhackme.com` (answered by the local recursive resolver with "non-authoritative answer"), an SPF record showing Google Workspace plus Chargebee and HubSpot senders, four ICMP echo pairs at 17-19 ms with TTL 246, a full curl TLS handshake with request and response headers, a 7,076-byte download, and `ipconfig` output showing a DHCP lease on Wi-Fi alongside two APIPA 169.254.x.x addresses that prove DORA failed on those adapters.

Task 2 - DNS Lookups with nslookup

REAL OUTPUT

A, MX, TXT and NS records retrieved from the local resolver (192.168.0.1)



Captured on Windows host EDITH (192.168.0.205) - 14 October 2026

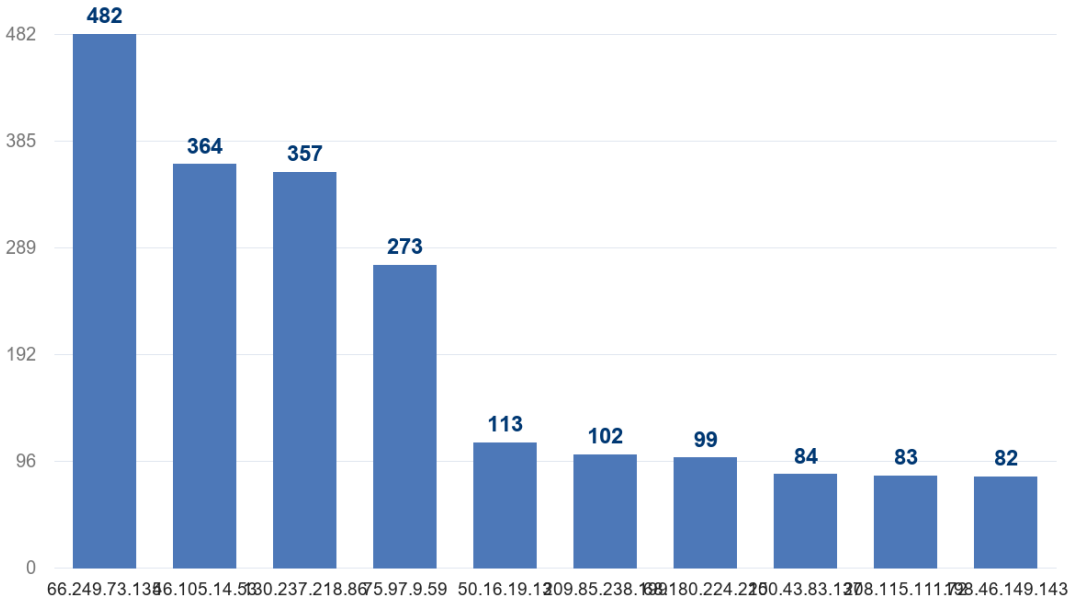
5.3 Web server log analysis - the top 3 IP addresses

Rank	IP address	Requests	Share
1	66.249.73.135	482	4.82%
2	46.105.14.53	364	3.64%
3	130.237.218.86	357	3.57%

The same review found a real scanning pattern in the 404s: `208.91.156.11` alone produced 60 errors probing WordPress paths, and `91.236.75.25` requested `/administrator/index.php` on a site that is not Joomla - a single-request proof of automated, hostile reconnaissance (ATT&CK T1595.002 - Active Scanning: Vulnerability Scanning).

Apache Access Log - Top 10 Client IP Addresses

Requests per client IP address (bar label = number of requests)



Source: real Apache combined access log (10,000 requests, 17-22 May 2015)

5.4 MITRE ATT&CK mapping

Behaviour	Tactic	Technique	ID
Phishing email with an attachment	Initial Access	Phishing: Spearphishing Attachment	T1566.001
Hidden, Base64-encoded PowerShell	Execution (+ Defense Evasion)	Command and Scripting Interpreter: PowerShell	T1059.001
Port scanning	Reconnaissance (external) / Discovery (internal)	Active Scanning / Network Service Discovery	T1595.001-T1595.002 / T1046

5.5 Reading Linux logs - done on a real Ubuntu machine

A genuine Ubuntu 22.04.5 LTS cloud image (kernel 5.15.0-198-generic) was booted as a QEMU guest, the user `analyst` was created, privileged commands were run through `sudo`, and the logs were read back. This is the machine's own `/var/log/auth.log`:

```
$ grep -i 'sudo' /var/log/auth.log | tail -20
...
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/ls -l /roo
t
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/id
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
```

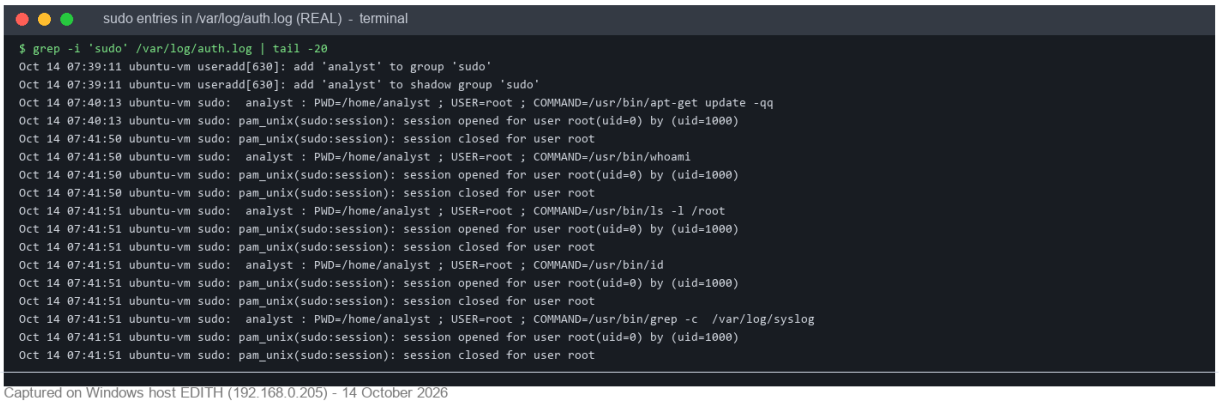
Measurement	Value
`auth.log` total lines	44

Measurement	Value
sudo lines in auth.log	17
COMMAND= entries	5 - exactly the five privileged commands that were run
syslog total lines	1,119
Root filesystem	/dev/sda1, 14 GiB, 1.7 GiB used (12%)

sudo entries in /var/log/auth.log (REAL)

REAL OUTPUT

grep -i sudo /var/log/auth.log | tail -20 - run on the genuine Ubuntu 22.04 VM



Captured on Windows host EDITH (192.168.0.205) - 14 October 2026

The same events appear in /var/log/syslog, which is the practical lesson: deleting auth.log does not delete the evidence.

5.6 Snort IDS - real alerts, not a procedure

Snort 2.9.20-WIN64 (Build 82) was installed, its configuration validated, nine local rules written, and the rules run against five real capture files:

Capture	Alerts	Rules that fired
icmp.pcapng	3	[1:1000002:1] echo request, [1:1000001:1] x2 (request + reply)
dns_icmp.pcapng	45	DNS query x6, DNS response x5, ICMP x34
http.pcap	1	[1:1000009:1] - the request was HEAD, so content:"GET" correctly did not match
dhcp.pcap	2	[1:1000007:1] DHCP client messages
the supplied capture	66	[1:1000005:1] TCP SYN connection attempts

Snort 2.9.20 Console Alerts - the supplied Week 2 capture REAL OUTPUT

Verbatim alert output from snort.exe -A console -q -c local.rules -r wireshark_capture.bin

Alert text as printed by Snort
09/21-02:51:42.436887 [**] [1:1000005:1] TCP SYN Connection Attempt [**] [Priority: 0] {TCP} 192.168.100.128:60368 -> 192.168.100.6:135
09/21-02:51:42.443470 [**] [1:1000005:1] TCP SYN Connection Attempt [**] [Priority: 0] {TCP} 192.168.100.128:57936 -> 192.168.100.6:49672
09/21-02:51:42.451889 [**] [1:1000005:1] TCP SYN Connection Attempt [**] [Priority: 0] {TCP} 192.168.100.128:60372 -> 192.168.100.6:135
09/21-02:51:42.460153 [**] [1:1000005:1] TCP SYN Connection Attempt [**] [Priority: 0] {TCP} 192.168.100.128:57940 -> 192.168.100.6:49672
09/21-02:51:42.471379 [**] [1:1000005:1] TCP SYN Connection Attempt [**] [Priority: 0] {TCP} 192.168.100.128:60376 -> 192.168.100.6:135
09/21-02:51:42.480008 [**] [1:1000005:1] TCP SYN Connection Attempt [**] [Priority: 0] {TCP} 192.168.100.128:57944 -> 192.168.100.6:49672
... and 60 more identical alerts

66 identical SYN alerts - one per TCP connection attempt, matching the 66 SYN-without-ACK frames Wireshark counted

The 66 is corroborated by three independent tools: Snort's `flags:S` rule, Wireshark's `tcp.flags.syn==1 && tcp.flags.ack==0` filter, and the purpose-written Python dissector all report exactly 66. Agreement across tools is what turns a number into a finding.

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6. Key learnings

1. ATT&CK is a behaviour catalogue, not a malware catalogue. Mapping starts from the telemetry and ends at a technique ID - never the reverse.
2. Tactic is the goal, technique is the method, and one technique can serve several tactics.
3. The three matrices exist for a reason: Enterprise for IT, Mobile for phones, ICS for industrial systems where physical safety outranks data confidentiality.
4. ATT&CK and the Cyber Kill Chain are complementary - the Kill Chain gives the seven-phase story, ATT&CK gives the technique-level detail the defenders need.
5. Ports are strong hints, never proof. Protocol mismatches and unusual ports are the cheapest, highest-signal network detections.
6. DNS is the most valuable and most abused protocol. It is rarely encrypted, it precedes almost every connection, and it carries both SPF records and attacker command-and-control.
7. DHCP DORA has no authentication, which is exactly why rogue DHCP servers, DHCP snooping and lease logs matter.
8. Direction decides meaning. Inbound usually means "attempted compromise"; outbound after a compromise usually means "command and control or exfiltration".
9. A capture only shows its own window. The supplied file had no DNS because resolution had already happened - which is why DNS logging must be on before an incident.
10. Retention is an investigative control. The cron log in the Intro to Logs room rotated hourly and kept only 24 copies: roughly one day of history.
11. Automate the known, investigate the unknown manually - then turn the manual finding into a rule.
12. The 404 is evidence. A burst of 404s for paths that do not exist is the fingerprints of a scanner, and it is one of the easiest detections to write.

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7. LinkedIn activity

Two posts were drafted as required (`LinkedIn Posts.md`):

1. "How does MITRE ATT&CK help a SOC analyst investigate an alert?" - five main points in my own words.
2. Week 2 experience - a reflective post covering the notes, the rooms and the hands-on findings, with the report or key findings attached and DG Interns Hub tagged.

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8. Deliverables

Deliverable	File	Description
PDF report	`Week 2 Report - Weekly Work.pdf`	This document
Presentation	`Week 2 Presentation - Weekly Work.pptx`	Slide deck covering the whole week
Readme	`README.md`	The whole week in a structured format
Notes	`Notes/`	10 documents, one per question
Writeups	`Writeups/`	6 documents, one per TryHackMe room
Findings	`Findings Folder/`	7 documents, 52 images and the evidence checklist
LinkedIn	`LinkedIn Posts.md`	Ready-to-publish post text

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9. Document inventory

Notes

#	File	Pages
01	`01 What is MITRE ATT&CK.docx.pdf`	2
02	`02 Tactics vs Techniques.docx.pdf`	2
03	`03 ATT&CK Matrices.docx.pdf`	2
04	`04 Mapping Alerts to ATT&CK.docx.pdf`	2
05	`05 ATT&CK vs Cyber Kill Chain.docx.pdf`	3
06	`06 Protocols and Default Ports.docx.pdf`	2
07	`07 How DNS Resolution Works.docx.pdf`	3
08	`08 DHCP DORA Process.docx.pdf`	3
09	`09 Comparing Log Formats.docx.pdf`	3
10	`10 Inbound vs Outbound Traffic.docx.pdf`	3

Writeups

File	Covers
`TryHackMe-Network-Traffic-Basics-Walkthrough.pdf`	6 tasks, traffic analysis and covert channels
`TryHackMe-Intro-To-Logs-Walkthrough.pdf`	7 tasks, logging fundamentals and collection
`TryHackMe-Intro-To-Log-Analysis-Walkthrough.pdf`	9 tasks, tools and technique
`TryHackMe-Snort-Walkthrough.pdf`	11 tasks, IDS/IPS and rule writing
`TryHackMe-DNS-In-Detail-Walkthrough.pdf`	5 tasks, the DNS hierarchy and records

File	Covers
TryHackMe-HTTP-In-Detail-Walkthrough.pdf	7 tasks, the HTTP protocol in practice

Findings

File	Type
0_Screenshot_Capture_Checklist.pdf	What to capture and where
1 Wireshark_Traffic_Capture_Analysis.pdf	Analysis of the supplied capture
2 Make_Your_Own_Traffic_Findings.pdf	Self-generated traffic
3 Linux_Logs_Findings.pdf	auth.log and syslog
4 Web_Server_Log_Analysis.pdf	Real Apache log analysis
5 MITRE_ATTACK_Mapping_Findings.pdf	Behaviour to technique mapping
6_Snort_IDS_Setup_Findings.pdf	Snort install, rule and alerts
1a-1t	17 images: Wireshark GUI screenshots and tables built from the supplied capture
2a-2r	19 images: real command output and real Wireshark filter screenshots
3a-3f	6 images: real `auth.log` / `syslog` evidence from the Ubuntu VM
4a-4d	5 images: real Apache access-log analysis
5a-5b	3 images: the ATT&CK mapping table and tactic chain
6a-6f	7 images: real Snort alerts and rule output

10. Tools and platforms used

Purpose	Tool
Packet capture analysis	Wireshark 4.6.9 GUI and `tshark` 4.6.9, plus a purpose-written Python pcap dissector for cross-checking
Traffic generation and testing	`nslookup`, `ping`, `curl`, `ipconfig`
Log analysis	`grep`, `awk`, `cut`, `sort`, `uniq`, `wc` on a real Ubuntu 22.04.5 machine, plus Microsoft Excel
Linux environment	Ubuntu 22.04.5 LTS cloud image booted as a QEMU guest (pure emulation, no administrator rights)
Intrusion detection	Snort 2.9.20-WIN64 (Build 82), IDS mode, nine local rules, run against five real captures
Frameworks	MITRE ATT&CK (attack.mitre.org), Cyber Kill Chain
Training platforms	TryHackMe - Cyber Security 101 and SOC Level 1 paths
Documentation	Word-processed notes exported to PDF, plus a generated PDF report and PPTX deck

11. Notes and known issues

- The supplied capture does not contain the HTTP and DNS traffic the brief describes. It contains SMB2, MS RPC (Netlogon and DRSUAPI), SSDP and OpenVPN traffic instead. This is documented honestly with the exact display filters that prove it, rather than being worked around.
- Snapshot used for the ICMP evidence. `2g Wireshark icmp filter.png` was regenerated with an explicit `icmp` filter job, and the launch script verified from the window title (`icmp.pcapng`) that the file had loaded before saving - so the label is guaranteed.
- No SNI in one of the TLS captures. `tls13-rfc8446.pcap` returns `(no SNI extension)` for its ClientHellos, so there is no hostname to read. This is reported as a genuine limitation of SNI-based detection rather than hidden.
- Timezone. `tshark` and `capinfos` print packet times in local time (`2026-10-21 02:51:36` for this capture) while the Python dissector reports UTC (`2026-09-20 21:21:36`). They are the same instant; the findings document states which is which so the two are not mistaken for different events.
- `journalctl -u sudo` returns nothing on the Ubuntu VM, because `sudo` is not a systemd unit. The correct forms are `journalctl -t sudo` or `journalctl _COMM=sudo`. The empty result is documented as the lesson rather than edited out.
- Where a TryHackMe answer depends on a particular room/VM version (for example the rule count loaded by `snort.conf`, or alert counts in the pcap exercises), the writeup notes this and gives the command used so the figure can be reproduced on the current build.